📌 What is SQL JOIN?

SQL JOIN is used to **combine rows from two or more tables** based on a related column (usually a foreign key).

👉 Helps retrieve connected data stored in different tables.

📚 Types of SQL Joins

1. INNER JOIN
2. LEFT JOIN
3. RIGHT JOIN
4. FULL JOIN
5. NATURAL JOIN

🧩 Example Tables

👨 Student Table

- ROLL_NO
- NAME
- AGE

📘 StudentCourse Table

- ROLL_NO
- COURSE_ID

Common Column: ROLL_NO

1️⃣ SQL INNER JOIN

✅ Definition:

Returns **only matching rows** from both tables.

Non-matching rows are excluded.

✅ Syntax:

```
SELECT columns

FROM table1

INNER JOIN table2

ON table1.column = table2.column;
```

👉 JOIN = INNER JOIN

✅ **Example:**

```
SELECT StudentCourse.COURSE_ID, Student.NAME, Student.AGE

FROM Student

INNER JOIN StudentCourse

ON Student.ROLL_NO = StudentCourse.ROLL_NO;
```

✓ **Result:**

Only students who are enrolled in courses.

2️⃣ SQL LEFT JOIN (LEFT OUTER JOIN)

✅ **Definition:**

- Returns **all rows from left table**
- Matching rows from right table
- If no match → NULL values

✅ **Syntax:**

```
SELECT columns

FROM table1

LEFT JOIN table2

ON table1.column = table2.column;
```

✅ **Example:**

```
SELECT Student.NAME, StudentCourse.COURSE_ID

FROM Student

LEFT JOIN StudentCourse

ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

✓ **Result:**

All students
Students without course → COURSE_ID = NULL

3️⃣ SQL RIGHT JOIN (RIGHT OUTER JOIN)

✅ **Definition:**

- Returns **all rows from right table**

- Matching rows from left table
- If no match → NULL values

✅ **Syntax:**

```
SELECT columns
FROM table1
RIGHT JOIN table2
ON table1.column = table2.column;
```

✅ **Example:**

```
SELECT Student.NAME, StudentCourse.COURSE_ID
FROM Student
RIGHT JOIN StudentCourse
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

✓ **Result:**

All course records
Courses without matching student → NAME = NULL

4️⃣ **SQL FULL JOIN (FULL OUTER JOIN)**

✅ **Definition:**

- Combines LEFT JOIN + RIGHT JOIN
- Returns **all rows from both tables**
- Non-matching rows → NULL values

✅ **Syntax:**

```
SELECT columns
FROM table1
FULL JOIN table2
ON table1.column = table2.column;
```

✅ **Example:**

```
SELECT Student.NAME, StudentCourse.COURSE_ID
FROM Student
FULL JOIN StudentCourse
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

✔ **Result:**

- All students
- All courses
- Matching rows combined
- Non-matching rows contain NULL

5️⃣ SQL NATURAL JOIN

✅ **Definition:**

- Automatically joins tables using **columns with same name**
- Works like INNER JOIN
- Common column appears only once

✅ **Example Tables:**

Employee:

- Emp_name
- Dept_id

Department:

- Dept_id
- Dept_name

✅ **Example Query:**

```
SELECT Emp_name, Dept_name
FROM Employee
NATURAL JOIN Department;
```

✔ **Result:**

Employees with their department names

🔥 Join Comparison Table

JOIN Type	Returns Matching	Returns Non-Matching
INNER JOIN	✅ Yes	❌ No
LEFT JOIN	✅ Yes	✅ Left Table Only

JOIN Type	Returns Matching	Returns Non-Matching
RIGHT JOIN	✅ Yes	✅ Right Table Only
FULL JOIN	✅ Yes	✅ Both Tables
NATURAL JOIN	✅ Yes	❌ No

🎯 Visual Understanding

Assume:

Table A = Students
Table B = StudentCourse

Join Type	Result
INNER	Only common data
LEFT	All Students + matching courses
RIGHT	All Courses + matching students
FULL	All Students + All Courses

🧠 Important Points

✓ JOIN is used with ON condition
✓ Foreign key relationships are commonly used
✓ FULL JOIN not supported in some databases (like older MySQL versions)
✓ NATURAL JOIN automatically matches common column names
✓ LEFT JOIN is most commonly used in real projects